Guppy on Quantinuum Systems Application

Please complete this form, providing as much detail as possible in your responses. The Quantinuum team will review your application and contact you with the next steps.

Applications submitted by Monday, September 7, 2026, will receive full consideration. Following the application deadline, the Quantinuum team will review all complete applications and notify applicants within approximately three weeks. Applications may continue to be accepted after this date, subject to program capacity. For questions, contact Q-Net@quantinuum.com.

Eligibility

- Applicants must be verified members of the Q-Net community. If you are not already a member, you can apply for free Q-Net membership by visiting https://www.quantinuum.com/q-net. Q-Net membership must be verified by the Quantinuum team before your application can be considered.
- Applicants should have a defined, time-bound project or workload that they intend to evaluate using Guppy on System Model H2.
- Experience with Guppy is not required. We welcome both new users and those migrating existing workloads from other quantum SDKs.
- Applicants must be willing to provide feedback on Guppy, onboarding, documentation, and the overall developer experience if selected for the program.

Acknowledgement: By submitting this application, applicants acknowledge that they have reviewed the **Guppy on Quantinuum Systems Access Program** details available **here** and understand that participation is subject to the program terms and conditions.

Answers aren't saved until submitted. Please copy

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Fields marked with an asterisk (*) are required.

Applicant Name: *

First Name, Last Name

Joseph, Castillo

Email Address: *

Use an email address associated with your organization

joseph.castillotorres@ucr.ac.cr

Participation requires Q-Net membership which is free. Q-Net membership status of primary applicant: *

Q-Net membership application submitted

Organization Name: *

University of Costa Rica

Department, Laboratory, or Research Group: *

If not applicable, put N/A

School of Physics

Organization Physical Address: *

Street address, city, state/province, postal code, and country

School of Physics, University of Costa Rica, San José 2060, Costa Rica

Is your organization a startup? *

No

Organization Website: *

Please enter a URL

https://www.fisica.ucr.ac.cr/

Organization Type: *

Academic Institution

Primary R&D Themes of your organization *

Select all that apply

Quantum Error Correction

Chemistry and Materials Science

Condensed Matter

List All Individuals Expected to Participate in the Project

Name | Organization | Email Address

Example:

Jane Smith | University of Example | janesmith@example.edu

Joseph Castillo Torres | University of Costa Rica | joseph.castillotorres@ucr.ac.cr; José Guillermo Fernández Blanco | University of Costa Rica | jose.fernandezblanco@ucr.ac.cr; Diego Hurtado Rojas | University of Costa Rica | diego.hurtadorojas@ucr.ac.cr; Guillermo Escobar Arrieta | guillermo.escobar@ucr.ac.cr; Kevin Martínez Herrera | Technical University of Munich | go59kah@mytum.de; Mauricio Gutiérrez Arguedas | University of Costa Rica | mauricio.gutierrez_a@ucr.ac.cr

Current Quantinuum tools in use *

Select all that apply

Guppy

pytket

Nexus

Current level of Quantinuum platform access" *

Select all that apply

No current access

Have you previously used Quantinuum's quantum computing hardware? *

No

Current Nexus organization name, if applicable:

-

Quantum SDKs or frameworks currently used: *

Select all that apply

Guppy

pytket

Qiskit

Other

If you selected other for Quantum SDKs or frameworks used, provide details: *

Stim

Briefly describe your experience with these tools: *

We have used these tools in the context of a quantum hackathon, where we focused on simulating the Ising and Fermi-Hubbard models. More specifically:

1) Guppy: We used one of Guppy's methods to simulate a Trotterized time evolution (quench with guppylang).

2) Pytket: We used extensively pytket, in particular we built and ran all the quantum circuits used in our simualtion of the Ising and the 2D Fermi-Hubbard models with pytket. This included error correction and error mitigation circuits.

3) Qiskit: For the hackathon we did preliminary explorations with qiskit. We have also used Qiskit to implement VQE circuits and other simple algorithms. We also have used Qiskit to implement QKD.

4) Stim: We have extensively used Stim to study and simulate QEC codes.

Project title: *

Quantum error corrected simulation of non-frustrated and frustrated spin systems

Technical Project Area *

Select all that apply

Quantum Error Correction

Fault-tolerant Quantum Computing

Quantum Simulation

Condensed Matter

Project summary: *

Provide a detailed description of the proposed project

The time evolution of spin systems is relevant to understand phenomena like magnetism and superconductivity. A digitized implementation of a continuous time evolution unavoidably bumps into the possibility of chaotic dynamics for long Trotter steps and extended evolution times. Recently, it has been observed that with Quantinuum's high-fidelity gates, it is possible to extend the evolution of the Ising model to long times [1].

This previous work was implemented with bare physical qubits. Our proposal aims to determine whether using encoded logical qubits allows for even longer time evolutions without the descent into chaotic regimes. More specifically, we plan to use various quantum error-detecting and correcting codes (Iceberg d=2, concatenated Iceberg d=4, 5-qubit code d=3, Steane d=3) to implement a Trotterized time evolution of the transverse-field Ising model (TFIM) and compute relevant observables (total magnetization, magnetization variance, two-point correlations, etc.) at different times. We will compare the time evolution using bare physical qubits with error mitigation techniques such as zero-noise extrapolation (ZNE) with full ED and EC protocols. We hypothesize that employing QEC will allow us to reach longer time evolutions.

We will explore square lattices as well as triangular lattices that exhibit frustration. We will also explore how accesible 3D lattices simulations become under this error-protected approach.

[1] R. Haghshenas et al, "Digital quantum magnetism on a trapped-ion quantum computer", Nature 653, 56-62 (2026)

Why does this work matter? *

Despite still being relatively small compared to future large-scale quantum computers, Quantinuum's high-fidelity hardware opens up the possibility of studying exciting phenomena in condensed-matter

models that lie at the frontier of what the best classical methods allow for. In particular, it has been shown it is possible simulate the time evolution and thermalization of spin systems for long time evolutions without descending into the chaotic regime.

Notably absent from this previous work is the use of quantum error correction to further lengthen the possible time evolution. QEC could offer a crucial boost to the fidelity of the logical gates, which could in turn allow for even longer time evolutions. Therefore, it is critical to determine whether or not the QEC codes implementable in the current quantum hardware would allow for this.

Frustrated systems are particularly interesting because competing interactions can create highly constrained dynamics, emergent excitations, and collective behavior. These features make them a stringent test of whether error protection can preserve subtle many-body physics that may be lost as errors accumulate. Recent works on Quantinuum H2 demonstrated thermalization and constrained dynamics in frustrated triangular spin systems, providing a direct benchmark for this project.

Suppressing errors could allow these systems to be evolved for longer times, making it possible to better resolve processes such as thermalization and relaxation and to study their emergent behavior more reliably.

Proposed methodology and technical approach: *

We will simulate the TFIM using second order Suzuki-Trotter decomposition, increasing lattice geometry and logical encoding progressively, such that each stage validates the next one. The first stage consists of unencoded square-lattice dynamics to establish state preparation, Trotter evolution, and observables measurement. Then, the project follows with encoded simulations in square lattices, and then to frustrated triangular lattices and restricted sector transport. This section will be the main focus of the project. Finally, Guppy will be implemented, mainly for mid-circuit discard, along with three dimensional simulations.

Observables include Ising-order fluctuations, spin correlations, energy-related quantities and their relaxation, and, on the frustrated lattice, dimer correlations, topological winding-number sectors, and constraint violations. For the non-frustrated lattice, results will be compared to known measurements on physical qubits. Transport will be characterized from controlled local perturbations by tracking defect density and the evolution of the system. Because constraint violations are measured in the same shots as the physics, and the emergent Gauss law is violated by the transverse field only at a small classically computable rate, the excess violation rate additionally provides an in-situ noise diagnostic at no extra cost.

The workflow will involve classical validation and circuit benchmarking, mainly through pytket and Guppy for dynamically controlled logical circuits.

On the error-corrected qubits, the logical gates associated with the Trotter steps will not be fully fault-tolerant since we will not have enough qubits to do a magic state distillation or any other FT protocol. Instead, we will employ partial FT and post-selection to achieve a sweet spot between low fidelity and efficiency.

Relevant prior work, preliminary results, code, or publications: *

Provide links where possible

Github repo of the hackathon challenge solved recently: https://github.com/guesar2/challenge3-hackathon; Google Scholar profile of the project PI: https://scholar.google.com/citations?user=XMla-DIAAAAJ&hl=en

Why is this project suitable for System Model H2 and Guppy? *

Describe the H2 capabilities you expect to use, why Guppy is an appropriate programming model for your workload, and any expected benefits compared with your current SDKs or development workflow.

Mid-circuit measurement and feedforward will be crucial to implement QEC. Mid-circuit measurement and reset will be used for QED. Also H2 will allow us to compare with the previous work performed on the same platform.

System Model H2 capabilities expected to be used: *

Select all that apply

Mid-circuit measurement and feedforward

Mid-circuit measure and reset

Anticipated software stack, programming languages, libraries, and dependencies: *

- Python: the project is written entirely in Python
- pytket: primary circuit construction and compilation
- pytket-quantinuum: H2 backend interface
- Qiskit: cross-validation of circuits and observables from prior work
- Guppy: programming layer for dynamically controlled logical circuits, expressing mid-circuit syndrome measurement and classical branching
- Quantinuum Nexus: job submission and management, access to the H2 emulator and System Model H2
- NumPy and SciPy: exact diagonalisation via sparse matrix representations and iterative eigensolvers, and general numerics, alongside Markov Chain Monte Carlo
- Stim: circuit-level stabiliser simulation to verify fault-tolerance properties of the encoding gadgets
- qermit: error mitigation for unencoded reference circuits
- pandas: data handling and analysis
- Bootstrap resampling: uncertainty estimation throughout

- Matplotlib: figures
- Git with pinned environments: version control and reproducibility across circuits, emulator runs, and hardware results

Current Guppy Readiness: *

Select the option that best describes the current state of your project

Concept Only

Describe any known technical risks, open questions, or dependencies:

There are several open questions: (1) since the Steane code needs a considerable number of qubits (7 data and 1-6 ancillary), will it allow us to implement a non-trivial lattice? (2) will the partially fault-tolerant protocols allow us to reach sufficiently low infidelities to simulate long time evolutions? (3) will the discarding rate of the error-detecting codes be low enough as to guarantee an efficient implementation?

What will you demonstrate first on the H2 emulator? *

First as a sanity check we will reproduce some of the results of the previous papers, then we will implement a small QEC code (Iceberg) to determine if allow us to extend the time evolution of the spin system. These tests involve lattices that can be simulated on the H2 emulator (less than 26 sites).

H2 emulator

What do you hope to accomplish on the H2 emulator before requesting access to H2 hardware? *

Describe the technical milestones or outcomes that you expect to achieve during the emulator stage.

(1) Be able to reproduce previous results

(2) Understand the limits of the QEC codes

(3) Built a full implementation that can be easily switch into the H2 hardware via a toggle.

How will you demonstrate that your workload is ready for H2 hardware? *

Describe the evidence you will provide (for example, validation results, performance measurements, benchmark comparisons, or other technical results) to support your transition from the emulator to hardware.

We will be ready to migrate to H2 hardware once we have validated our simulations by reproducing the previous paper's [1] results. All of the preliminaries results obtained during the validation with the emulator will be uploaded to our Github repo.

[1] R. Haghshenas et al, “Digital quantum magnetism on a trapped-ion quantum computer”, Nature 653, 56-62 (2026)

Estimated time required to complete emulator validation: *

2-3 weeks

Do you need a local emulator for System Model H2? *

Yes

Estimated number of qubits per circuit (if known):

55

Estimated number of two-qubit gates per circuit (if known):

2000

Estimated total Hardware Quantum Credits (HQCs) requested:

45000

For guidance, see the HQC documentation here

Primary project intent: *

Select all that apply

Academic research

Publication

Expected project outcomes: *

Select all that apply

Research results

Publication or preprint

Evaluation of Guppy developer experience

Feedback on documentation or onboarding

Are you willing to provide feedback on Guppy, onboarding, documentation, and the overall developer experience? *

Yes